

FAI Job Search Multi-Task Agent System

Complete Technical Architecture, Agent Sequencing, APIs & End-to-End Workflow Specification

1. Executive Overview & System Design

The system is a fully autonomous, failover-resilient multi-agent architecture built to transform raw resumes into highly targeted, ATS-optimized resumes aligned with real-time job openings. It orchestrates specialized AI agents across five distinct stages, supported by deterministic Python tools for parsing, live multi-portal scraping, and PDF generation.

2. Quick Technical Reference Matrix

Stage / Agent	Primary Task	Tool / Library	LLM / Data Provider
Stage 0 File Ingestion	Extract raw text from PDF	pypdf.PdfReader	Deterministic Local Tool (No LLM required)
Stage 1 ResumeParserAgent	Extract structured candidate ontology (Contact, Skills, Exp, Projects)	Structured JSON Schema CandidateProfile	OpenRouter → Groq → Gemini (Structured LLM Failover)
Stage 2 RoleRecommenderAgent	Goal Formulation & State Space Planning (Ranked roles with fit score)	Structured JSON Schema RoleRecommendationResponse	OpenRouter → Groq → Gemini (Structured LLM Failover)
Stage 3 JobSearchAgent	Live Multi-Portal Job Retrieval & Keyword/Location Filtering	JobSearchTool JobCache (SQLite) requests + bs4	5 Free Job Feeds: 1. LinkedIn Guest Scraper 2. Remote REST API 3. Jobicy Tech API 4. RemoteOK Feed 5. ArbeitNow Board
Stage 4 MatchGapAgent	Heuristic Evaluation & Semantic Skill Gap Analysis (Score 0-100%)	Structured JSON Schema MatchReport	OpenRouter → Groq → Gemini (Structured LLM Failover)
Stage 5A TailorAgent (Generator)	Rewrite summary, projects, and work bullets using STAR method	Structured JSON Schema TailoredResume	OpenRouter → Groq → Gemini (Structured LLM Failover)
Stage 5B VerifierAgent (Auditor)	Fact-check tailored resume against original to catch hallucinated facts	Structured JSON Schema VerificationReport	OpenRouter → Groq → Gemini (Structured LLM Failover)
Stage 6 PDF Generator	Compile ATS-compliant, styled 1-2 page tailored PDF	reportlab.platypus (Flowable Layout Engine)	Deterministic Local Tool (No LLM required)

3. Detailed Sequential Agent Execution Flow

Stage 0: Document Ingestion & Text Extraction

Trigger: User uploads a PDF resume or selects a pre-loaded sample resume in Streamlit.
Component: tools/pdf_reader.py via pypdf.PdfReader.
Operation: Iterates through binary document stream, strips header/footer anomalies, and normalizes UTF-8 strings.
LLM Cost: \$0 (Pure deterministic execution).

Stage 1: ResumeParserAgent (Ontology Extraction)

Trigger: Receives normalized resume text from Stage 0.
Component: agents/resume_parser_agent.py.
LLM Call: self.call_structured_llm(..., response_schema=CandidateProfile).
Failover Architecture: Automatically routes through OpenRouter (Free models: Llama 3.3, Gemma 27B) → Groq (Llama 3.3-70b-versatile) → Google Gemini (gemini-2.5-flash via google-genai SDK).
Output: Strongly-typed CandidateProfile object containing contact, education, categorized technical skills, experience, and inferred seniority level.

Stage 2: RoleRecommenderAgent (Goal Formulation)

Trigger: Receives structured CandidateProfile.

Component: agents/role_recommender_agent.py.

LLM Call: Evaluates candidate capabilities against industry roles to produce RoleRecommendationResponse.

Output: Ranked role titles (e.g. 'AI Engineer', 'Backend Developer') with percentage fit, strengths, and gap areas. The user selects their target role and location.

Stage 3: JobSearchAgent & Feed Tooling (Informed Search)

Trigger: User selects target role and preferred location (e.g. 'Hyderabad', 'India', 'Remote').

Components: JobSearchAgent → JobSearchTool → JobCache (SQLite).

Data Retrieval Strategy:

1. Checks local SQLite database (data/job_cache.db) for matching records < 24h old.
2. On cache miss or forced refresh, queries **5 free live job providers** concurrently:
 - **LinkedIn:** Public guest scraper (/jobs-guest/jobs/api/seeMoreJobPostings/search) for live regional & remote jobs.
 - **Remotive API:** Official REST endpoint (https://remotive.com/api/remote-jobs) with token filtering.
 - **Jobicy API:** Direct JSON feed (https://jobicy.com/api/v2/remote-jobs) for global tech listings.
 - **RemoteOK API:** Public developer feed (https://remoteok.com/api).
 - **ArbeitNow API:** European & international remote job board API.
3. Performs query expansion if initial recall is 0, normalizes payloads, and persists them into SQLite.

Stage 4: MatchGapAgent (Heuristic Evaluation & Gap Analysis)

Trigger: User selects target job listing.

Component: agents/match_gap_agent.py.

LLM Call: self.call_structured_llm(..., response_schema=MatchReport).

Output: MatchReport mapping exact matching skills, critical missing skill gaps, match percentage score (0-100%), and tailoring recommendations.

Stage 5: TailoringOrchestrator (Adversarial Generator-Verifier Loop)

Component: agents/tailoring_orchestrator.py.

Sub-Agent 5A (TailorAgent): Re-writes resume summary, technical skills, and project bullet points using the STAR format (Situation, Task, Action, Result) to organically highlight job keywords.

Sub-Agent 5B (VerifierAgent): Acts as an adversarial auditor. Checks the tailored resume against the original profile to verify that degrees, companies, and skill claims are strictly truthful without hallucinated facts.

Loop Termination: If verified (score ≥ 75), proceeds to generation. If hallucinations are detected, re-prompts TailorAgent with verifier critique notes (up to 2 iterations).

Stage 6: PDF Generation & Compilation

Component: tools/pdf_generator.py via reportlab.platypus.

Operation: Transforms verified TailoredResume data model into an ATS-friendly, clean, styled PDF with precise margins, typography, contact headers, and bullet hierarchy.

Final Deliverable: Stored in output_resumes/ and presented in Streamlit UI for immediate single-click download.